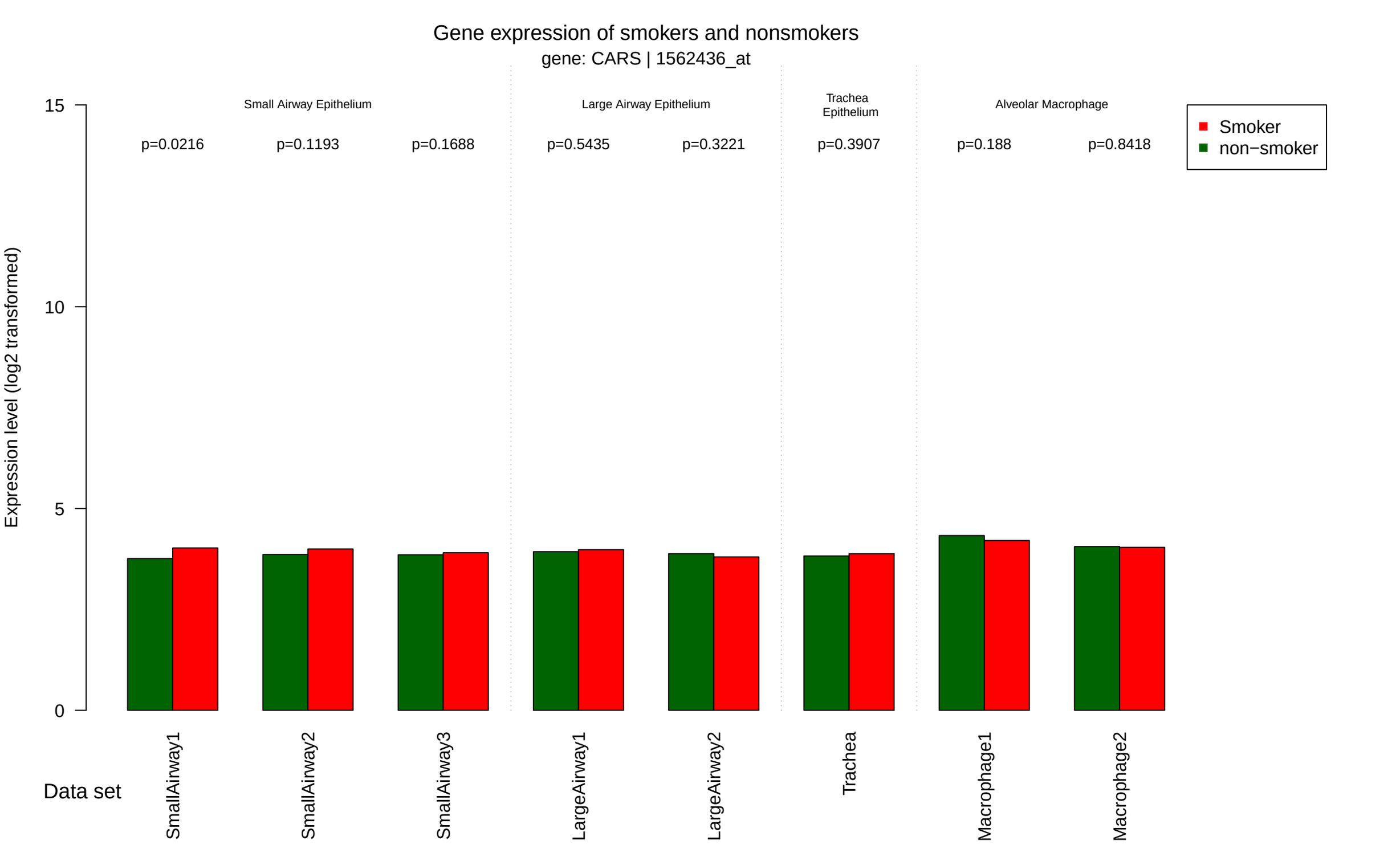
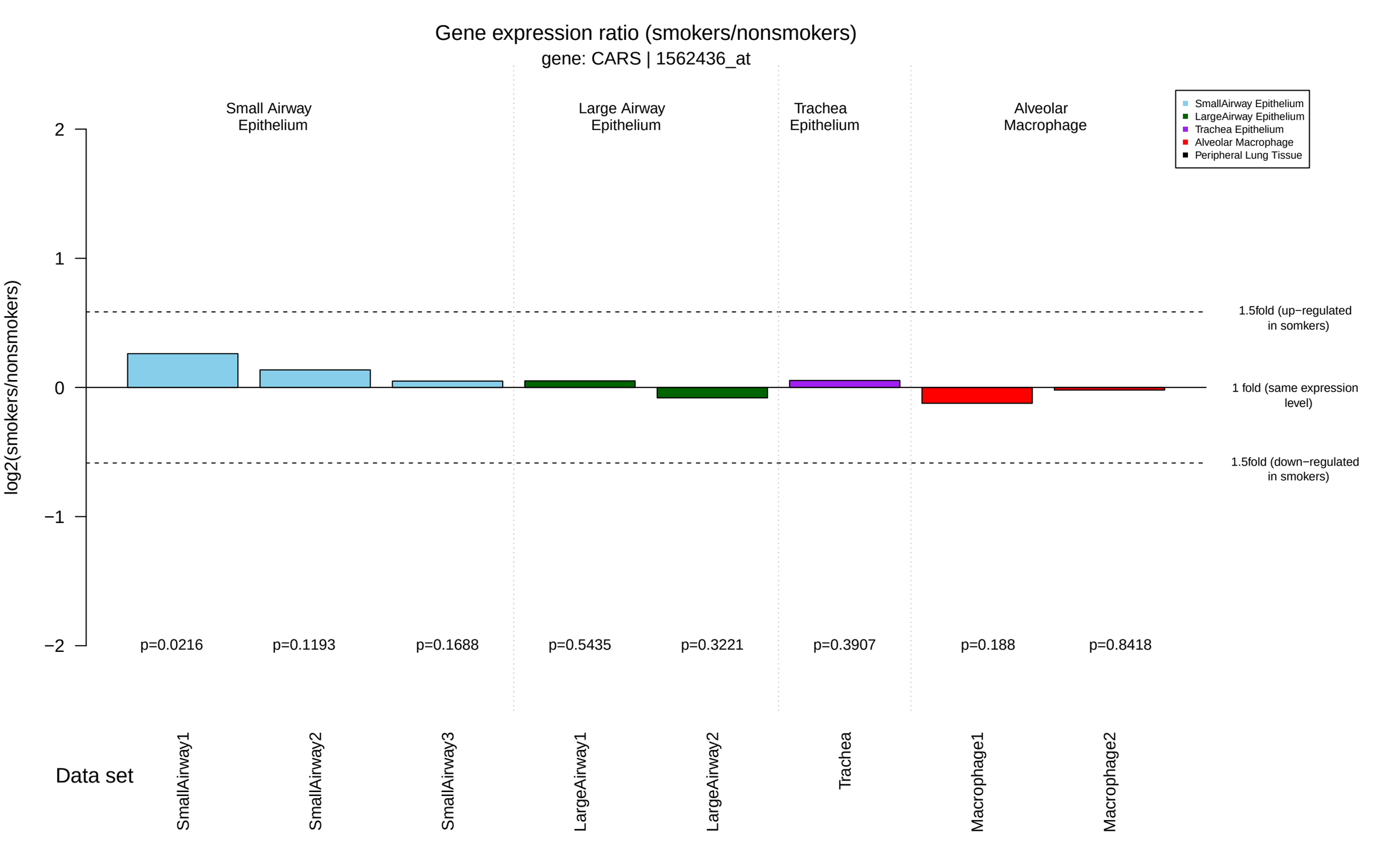
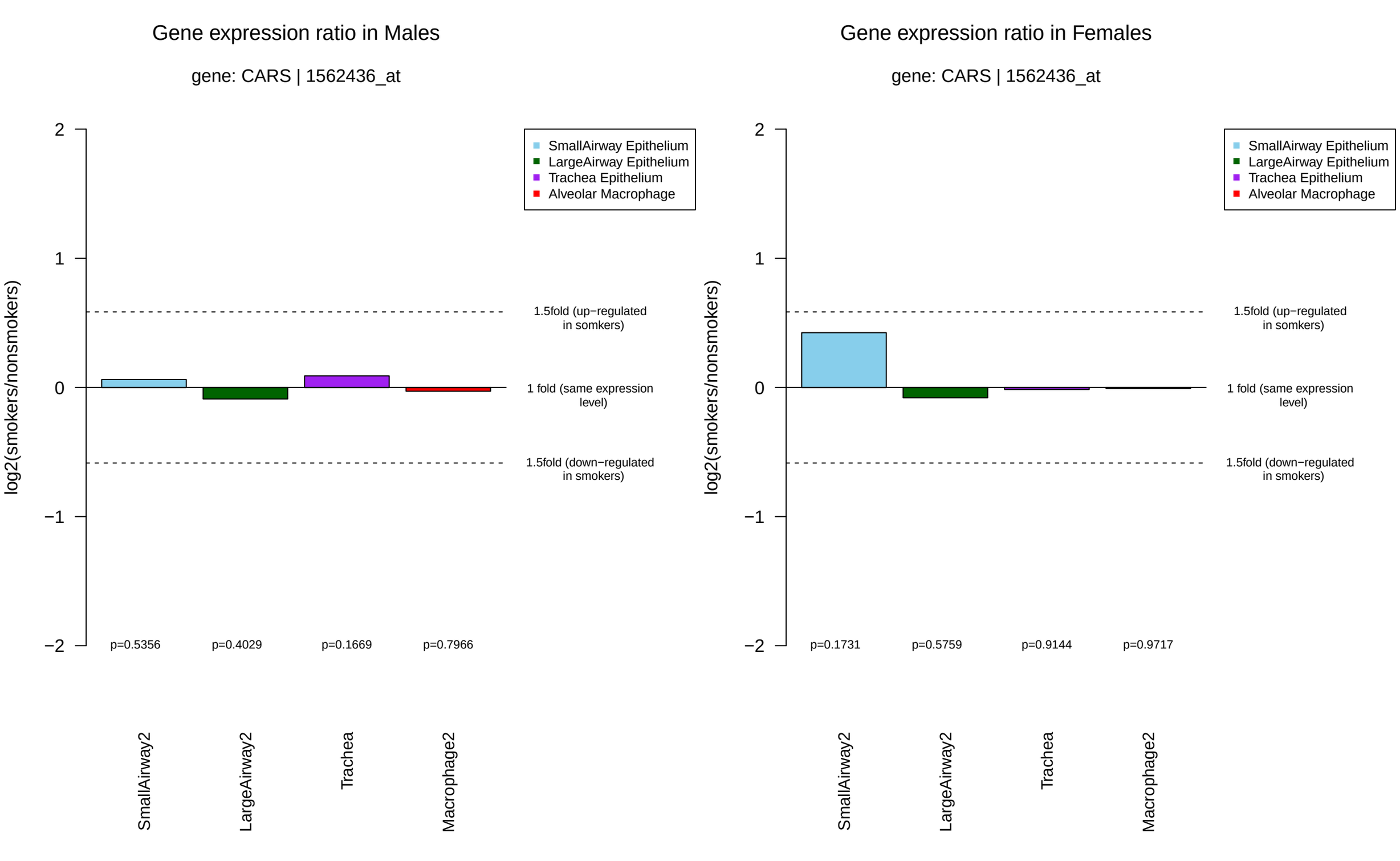
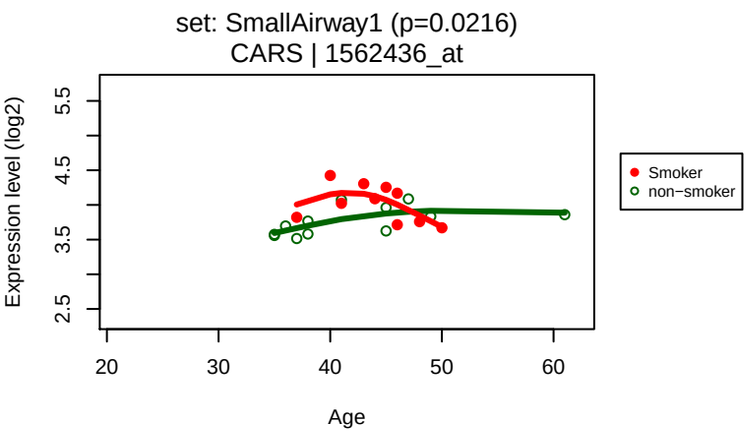
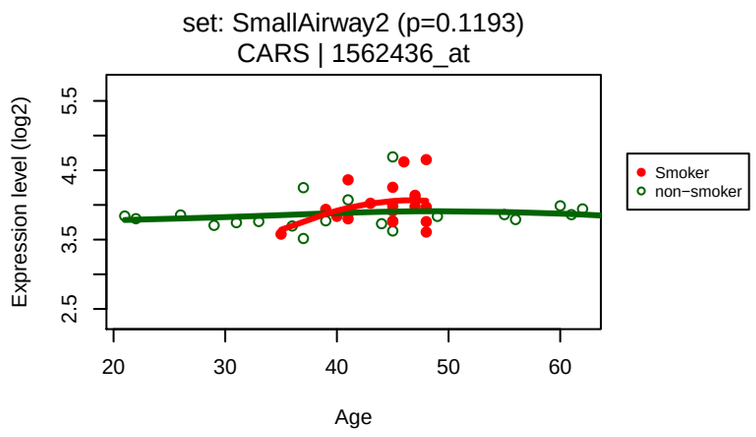
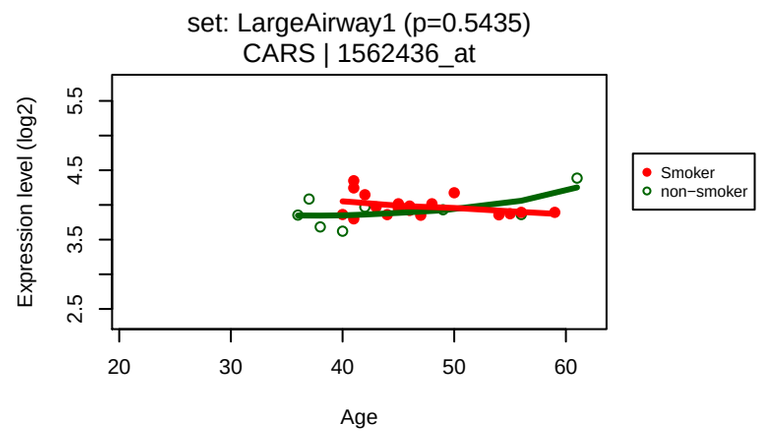

Gene expression of smokers and nonsmokers
gene: CARS | 1562436_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0216
p=0.1193
p=0.1688
p=0.5435
p=0.3221
p=0.3907
p=0.188
p=0.8418
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

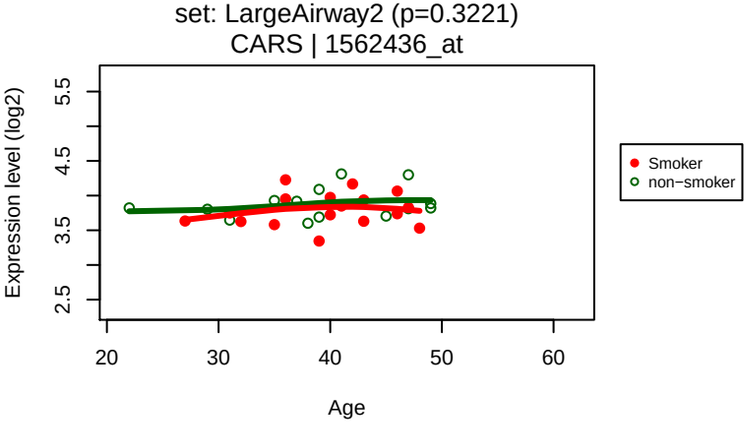

Gene expression ratio (smokers/nonsmokers)
gene: CARS | 1562436_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0216
p=0.1193
p=0.1688
p=0.5435
p=0.3221
p=0.3907
p=0.188
p=0.8418
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

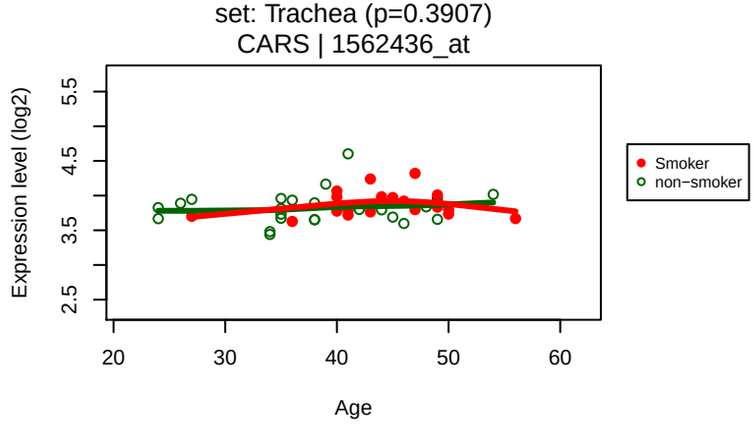

Gene expression ratio in Males
gene: CARS | 1562436_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5356
p=0.4029
p=0.1669
p=0.7966
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CARS | 1562436_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1731
p=0.5759
p=0.9144
p=0.9717
SmallAirway2
LargeAirway2
Trachea
Macrophage2

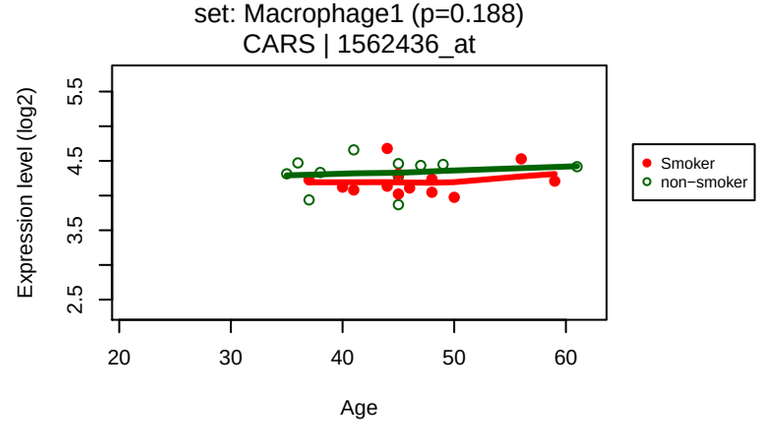

set: SmallAirway1 (p=0.0216)
CARS | 1562436_at
Expression level (log2)
Age
Smoker
non−smoker

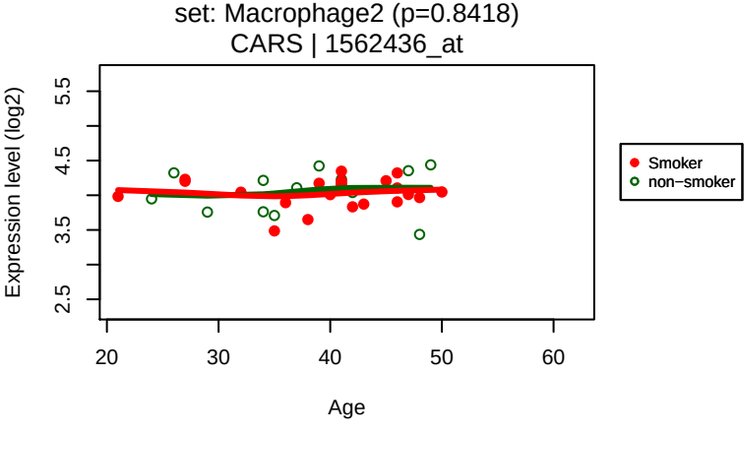

set: SmallAirway2 (p=0.1193)
CARS | 1562436_at
Expression level (log2)
Age
Smoker
non−smoker

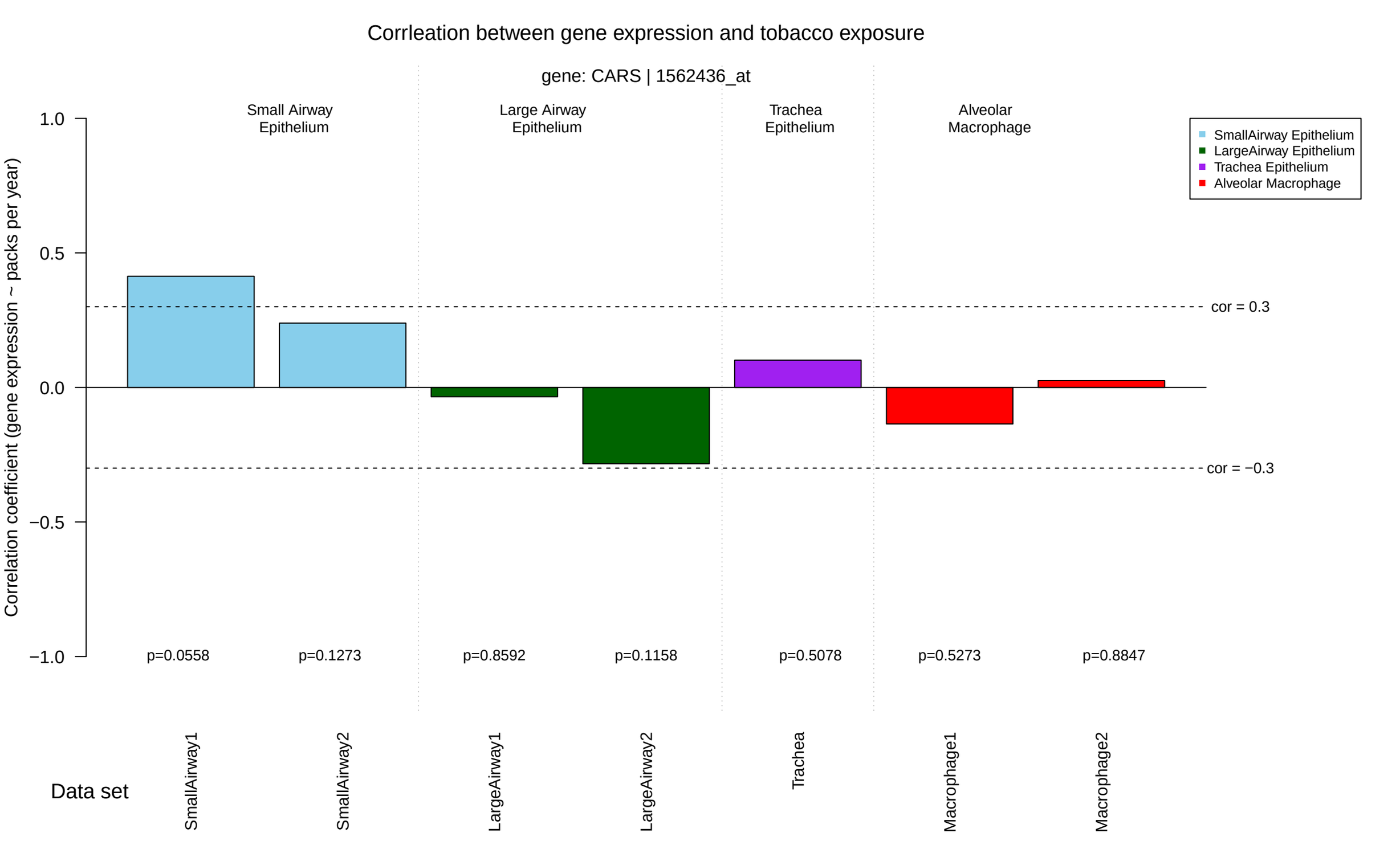

set: LargeAirway1 (p=0.5435)
CARS | 1562436_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3221)
CARS | 1562436_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3907)
CARS | 1562436_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.188)
CARS | 1562436_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8418)
CARS | 1562436_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CARS | 1562436_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0558
p=0.1273
p=0.8592
p=0.1158
p=0.5078
p=0.5273
p=0.8847
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2